

DARKTRACE: Experimental Methodology

Explainable and Forensically Verifiable Multilingual Dark Web Threat Intelligence Framework

1 Experimental Methodology

1.1 Design Philosophy and Overview

The evaluation is organized around six research targets: (T1) threat-text classification, (T2) multilingual detection, (T3) severity-scoring accuracy, (T4) explainability quality, (T5) evidence integrity (SHA-256 hashing and chain-of-custody), and (T6) comparison against baselines. Each target is treated as a separate, pre-registered experiment with its own hypotheses, metrics, and statistical tests, so that the integrated framework is justified component-by-component as well as end-to-end. All data are drawn from *sanitized, archived, public, or synthetic* sources; no live, illegal, or unauthorized collection is performed (Section 1.2). The pipeline, splits, seeds, and configurations are released for reproducibility (Section 1.14).

Hypotheses.

- H1** A fine-tuned multilingual transformer significantly outperforms classical ML and monolingual deep baselines on threat classification (T1, T6).
- H2** Multilingual joint training narrows the cross-lingual gap; zero-shot transfer incurs a bounded, measurable penalty relative to monolingual upper bounds (T2).
- H3** Hybrid severity scoring (classifier posterior + interpretable features) is better calibrated and better ranked than classifier-confidence and static rule baselines (T3).
- H4** SHAP and attention yield higher explanation fidelity than LIME on this text domain, and the three are complementary for analysts (T4).
- H5** The integrity layer achieves 100% tamper detection with 0% false accepts at low throughput overhead (T5).

1.2 Ethical Dataset Sources

We use only datasets that are publicly released for research, archived, sanitized, or synthetically generated. Table 1 summarizes the layered strategy. Raw illicit content is never re-published; only derived features, labels, and hashes are released where licensing permits. Personally identifying information (PII) and victim data are removed or pseudonymized before any processing.

Synthetic data protocol. To address low-resource languages and rare/harmful classes without collecting illegal material, we generate synthetic threat-style text from templates and instruction-tuned models, conditioned on *abstracted* threat patterns (never operational attack instructions). Every synthetic item is human-reviewed, labeled, and flagged as synthetic so its effect can be ablated. Synthetic data is used to *augment*, never to replace, the real evaluation sets, and all headline results are reported on real (non-synthetic) test data.

Table 1: Layered, ethical dataset strategy. “Role” indicates the target(s) each source serves. Verify each source’s licence and re-use terms before publication.

Layer / Source	Description	Role
Archived darknet corpora (e.g. sanitized Agora/forum archives) [23]	Publicly archived, de-identified marketplace listings and forum posts used in prior peer-reviewed work	T1, T3, T6
CTI NER corpora (APTNER, DNRTI, CyNER, CyberNER) [28, 8, 7, 9]	STIX-aligned, public annotated CTI text	actor/entity, T1
Open-source CTI alert datasets (e.g. CySecAlert) [20]	Public OSINT security-event text with relevance labels	T1, T6
Public vulnerability text (NVD/CVE descriptions)	Standardized severity-bearing descriptions for severity supervision	T3
Multilingual augmentation (machine-translated subsets + native low-resource crime text) [1, 18]	EN, HI, AR, RU, FR/DE evaluation sets	T2
Synthetic threat text (template- and LLM-generated, human-reviewed)	Controlled, fully shareable data for low-resource languages and rare classes	T2, T3

1.3 Threat Class Labels

We adopt a flat, mutually informative taxonomy of threat categories plus a non-threat class (Table 2), chosen to be coverage-complete for dark-web CTI while remaining annotatable with high agreement. Severity is modeled *orthogonally* as an ordinal band (Section 1.7), so a single category (e.g. “data leak”) can span severities.

Annotation. Two or more trained annotators label each item for category and severity following a written codebook; we report inter-annotator agreement (Cohen’s κ / Krippendorff’s α) and adjudicate disagreements. Class imbalance is expected and handled at training and evaluation time (macro-averaging, class-balanced loss).

1.4 Preprocessing

Preprocessing is identical across models to ensure fair comparison and is logged as hashed, derived artifacts (T5). Steps:

1. **Normalization:** Unicode/encoding normalization, HTML/boilerplate stripping, whitespace and case handling (case preserved for transformers).
2. **Noise handling:** removal of markup, navigation, and signature blocks; optional defanging of live IOCs (e.g. `hxxp`) without losing the token signal.
3. **Deduplication:** exact and near-duplicate removal via MinHash/SimHash to prevent train/test leakage from reposted content.
4. **Language identification:** per-document language tagging; routing to the native-multilingual path and, in parallel, a translate-then-classify path.

Table 2: Threat-category labels (classification) and severity bands (scoring).

Threat category	Definition (annotation guideline summary)
Malware / Ransomware	Offering, sharing, or discussing malicious code, builders, or ransomware operations
Exploit / Vulnerability	Sale or disclosure of exploits, 0-days, PoCs, CVE-linked weaknesses
Data Leak / Credential	Trading or dumping stolen credentials, databases, PII
Fraud / Carding	Payment-card fraud, money laundering, scam services
Hacking / Attack Services	Attack-as-a-service, botnets, DDoS-for-hire, access brokering
Other Illicit Trade	Non-cyber illicit goods/services (retained as a coarse class; may be excluded)
Benign / Non-threat	Off-topic, social, or non-malicious content (negative class)
Severity band	Criterion
Critical / High / Medium / Low	Ordinal scale from analyst rubric over impact, exploitability, target sensitivity, and recency/urgency

5. **Translation (parallel path):** machine translation of non-English text to English, retained as a separate derived artifact so its effect is ablatable.
6. **Tokenization/representation:** TF-IDF / n -grams for classical models; subword tokenization for transformers [5, 3].
7. **Feature extraction (severity):** interpretable threat features (CVE/exploit mentions, credential counts, target-sensitivity cues, urgency terms, actor-reputation signals) computed for the hybrid scorer.

1.5 Models

Baseline ML. TF-IDF features with Logistic Regression, Linear SVM, Random Forest, and XGBoost—fast, interpretable lower bounds and the T6 reference point.

Deep (non-transformer). CNN, BiLSTM, and BiLSTM-CRF (the CRF variant for sequence/NER tasks) [13].

Monolingual transformers. BERT/RoBERTa per language as per-language upper bounds [5, 23].

Multilingual transformers. mBERT and XLM-R (core classifier) [3, 27], evaluated under joint multilingual, zero-shot, and few-shot transfer regimes [26].

Generative/LLM baselines. An instruction-tuned LLM evaluated zero-/few-shot for classification and extraction [24, 25].

Severity head (proposed). A calibrated hybrid mapping {posterior, interpretable features} $\rightarrow s \in [0, 1]$ and an ordinal band, compared against (i) raw classifier confidence and (ii) a static rule/CVSS-style baseline [29, 21].

1.6 Train/Test Splits and Protocol

- **Primary split:** stratified 70/15/15 train/validation/test, preserving class proportions, with fixed seeds.
- **Cross-validation:** stratified 5-fold CV for classical and deep models; for transformers, 3 seeds $\times$ the held-out test set, reporting mean $\pm$ std.
- **Cross-lingual protocol (T2):** (a) *monolingual* (train/test same language) as upper bound; (b) *zero-shot* (train EN, test HI/AR/RU/FR/DE); (c) *few-shot* (k target-language examples added). We report the transfer gap $\Delta F1$.
- **Leakage control:** deduplication before splitting; near-duplicates and reposts forced into the same split; language-disjoint and (where timestamps exist) *temporal* splits (train on older, test on newer) to test robustness to drift.
- **Synthetic isolation:** synthetic data may enter training only; all headline test sets are real.

1.7 Severity Scoring Evaluation

Severity is evaluated as both *ranking* and *calibration*: ranking via NDCG@ k and Spearman/Kendall correlation against the analyst-assigned ordering; calibration via Expected Calibration Error (ECE), Brier score, and reliability diagrams, with temperature scaling applied post-hoc. Ordinal-band performance uses macro-F1 and Quadratic Weighted Kappa (which penalizes distant misclassifications more).

1.8 Explainability Evaluation

We compare SHAP [16], LIME [19], and transformer attention, following comparative-XAI practice in security [11, 10, 12, 22]:

- **Fidelity:** deletion/insertion and AOPC (area over the perturbation curve)—removing high-attribution tokens should drop confidence fastest for the most faithful method.
- **Stability/robustness:** attribution agreement under small input perturbations and across seeds.
- **Complexity/sparsity:** number of features needed to explain a decision.
- **Cross-method agreement:** SHAP–LIME–attention rank correlation.
- **Cross-lingual fidelity:** whether explanation quality degrades under translation/transfer (links T2 and T4).

1.9 Evidence Integrity Evaluation

The integrity layer is validated with an adversarial *tamper test*. A controlled set of post-hoc modifications—single/multi bit-flips, token substitution, artifact deletion, custody-record reordering, and ledger edits—is injected, and verification recomputes SHA-256 digests and re-validates the hash-chained custody log [15, 2, 14]. Metrics: tamper-detection rate (target 100%), false-accept rate (target 0%), verification latency, custody-log completeness, and the throughput overhead the layer imposes on the analytic pipeline.

1.10 Evaluation Metrics Summary

1.11 Ablation Study

We remove or alter one capability at a time and measure the effect on the most-affected axis (Table 4), establishing that *integration* is the contribution rather than any single module.

Table 3: Metrics by research target.

Target	Metrics
T1 Classification	Accuracy, macro-/weighted-F1, per-class precision/recall/F1, MCC
T2 Multilingual	Per-language F1, macro-F1 across languages, transfer gap Δ F1, language-wise confusion
T3 Severity	NDCG@ k , Spearman/Kendall ρ , ECE, Brier, Quadratic Weighted κ
T4 Explainability	Fidelity (AOPC), stability, complexity, cross-method agreement
T5 Integrity	Tamper-detection rate, false-accept rate, verification latency, log completeness, overhead %
T6 Baselines	All of the above, head-to-head, with statistical tests (Sec. 1.13)

Table 4: Ablation configurations (result cells filled from experiments).

Configuration	Primary axis	Δ vs. full
Full DARKTRACE	—	—
– multilingual (English-only)	T2 cross-lingual F1	—
– native vs. translate-then-classify	T2 cross-lingual F1	—
– hybrid features (posterior only)	T3 calibration (ECE)	—
– calibration (no temperature scaling)	T3 ECE/Brier	—
SHAP only / LIME only / attention only	T4 fidelity	—
– synthetic augmentation	T1/T2 low-resource F1	—
– integrity layer	T5 tamper detection	—
– STIX extraction / link analysis	actor recall / usability	—

1.12 Tables and Graphs to Include

Tables: (1) dataset summary with per-language/per-class counts; (2) main classification results (Acc/F1/MCC) across all models; (3) per-language and transfer-gap table; (4) severity ranking + calibration table; (5) explanation-quality table (fidelity/stability/agreement); (6) integrity tamper-test table; (7) ablation table; (8) statistical-test summary (p-values, effect sizes). **Figures:** (a) architecture/data-flow diagram; (b) per-class confusion matrices (overall and per language); (c) reliability diagram for severity calibration; (d) AOPC/deletion curves comparing SHAP/LIME/attention; (e) example SHAP/attention token-attribution heatmaps (EN and one non-EN language); (f) critical-difference (CD) diagram from the Friedman–Nemenyi test; (g) actor link-analysis graph; (h) integrity-overhead vs. throughput plot.

1.13 Statistical Validation

All comparisons report central tendency with dispersion and significance:

- **Pairwise classifier comparison:** McNemar’s test on the shared test set [17]; the 5×2 cv paired t -test for resampled comparisons [6].
- **Multiple models over multiple datasets/languages:** Friedman test with Nemenyi post-hoc and a critical-difference diagram [4].
- **Confidence intervals:** bootstrap (e.g. 1000 resamples) 95% CIs for F1, NDCG, and ECE.

- **Effect sizes and correction:** report effect sizes (e.g. Cliff’s δ) and apply Holm/Bonferroni correction for multiple comparisons.
- **Calibration significance:** compare ECE/Brier with bootstrap CIs across scoring variants.
- Significance threshold $\alpha = 0.05$ unless corrected.

1.14 Reproducibility Checklist

- ☐ Public release of code, configs, and trained-model hashes.
- ☐ Dataset datasheet; construction scripts and exact split files; SHA-256 of each released artifact.
- ☐ Fixed random seeds; reported hardware, library versions, and environment file.
- ☐ Hyperparameter search space and selected values; training/inference cost (time, energy).
- ☐ Exact preprocessing, tokenization, and translation settings.
- ☐ Per-run metrics (not just best run); mean $\pm$ std over seeds/folds.
- ☐ Statistical-test scripts and raw outputs.
- ☐ Ethics statement, data-management plan, and licence compliance for every source.
- ☐ Synthetic-data generation prompts/templates and human-review records.

1.15 Limitations

(1) Archived/sanitized and synthetic data may underrepresent the most recent, ephemeral threats; live coverage is constrained by ethics and law. (2) Severity labels are partly subjective; inter-annotator agreement bounds calibration claims. (3) Machine translation can distort low-resource semantics and the explanations derived from translated text—an effect we measure but cannot fully remove. (4) Synthetic augmentation risks distributional mismatch; we mitigate by testing only on real data and ablating synthetic contributions. (5) The permissioned-ledger integrity model resists post-hoc tampering but assumes trusted timestamping and does not prevent collection-time fabrication. (6) Adaptive adversaries (obfuscation, code-switching, slang drift) may evade detection, motivating the temporal-split robustness analysis.

References

- [1] Njood K. Al-harbi et al. Assessing BERT-based models for arabic and low-resource languages in crime text classification. *PeerJ Computer Science*, 2025.
- [2] Mohammad Ali et al. A procedure for tracing chain of custody in digital image forensics: A paradigm based on grey hash and blockchain. *Symmetry*, 14, 2022.
- [3] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- [4] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.
- [5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, 2019.

- [6] Thomas G. Dietterich. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7):1895–1923, 1998.
- [7] Yasir Ech-Chammakhy et al. CyberNER: A harmonized STIX corpus for cybersecurity named entity recognition. In *2025 IEEE 24th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2025.
- [8] Pavlos Evangelatos et al. Named entity recognition in cyber threat intelligence using transformer-based models. In *2021 IEEE International Conference on Cyber Security and Resilience (CSR)*, 2021.
- [9] Xinzhu Feng et al. PROMPT-BART: A named entity recognition model applied to cyber threat intelligence. *Applied Sciences*, 2025.
- [10] Diogo Gaspar et al. Explainable AI for intrusion detection systems: LIME and SHAP applicability on multi-layer perceptron. *IEEE Access*, 2024.
- [11] Pamela Hermosilla et al. Explainable AI for forensic analysis: A comparative study of SHAP and LIME in intrusion detection models. *Applied Sciences*, 15, 2025.
- [12] Rajesh Kalakoti et al. Evaluating explainable AI for deep learning-based network intrusion detection system alert classification. *Journal of Information Security and Applications*, 2025.
- [13] Gyeongmin Kim et al. Automatic extraction of named entities of cyber threats using a deep Bi-LSTM-CRF network. *International Journal of Machine Learning and Cybernetics*, 2020.
- [14] Leandro Loffi et al. Management of the chain of custody of digital evidence using blockchain and self-sovereign identities: A systematic literature review. *IEEE Access*, 2025.
- [15] Auqib Hamid Lone and Roohie Naaz Mir. Forensic-chain: Blockchain based digital forensics chain of custody with PoC in hyperledger composer. *Digital Investigation*, 28, 2019.
- [16] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [17] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- [18] Muhammad Kashif Nazir et al. Leveraging multilingual transformer for multiclass sentiment analysis in code-mixed data of low-resource languages. *IEEE Access*, 2025.
- [19] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. “why should i trust you?”: Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.
- [20] Thea Riebe, Tristan Wirth, Markus Bayer, Philipp Kuehn, Marc-André Kaufhold, Volker Knauthe, Stefan Guthe, and Christian Reuter. CySecAlert: An alert generation system for cyber security events using open source intelligence data. In *Information and Communications Security (ICICS)*, 2021.
- [21] Santosh Saklani et al. Severity prediction of software vulnerabilities using convolutional neural networks. *Information and Computer Security*, 2025.
- [22] Ahmed M. A. Salih et al. A perspective on explainable artificial intelligence methods: SHAP and LIME. *Advanced Intelligent Systems*, 2023.

- [23] Kanti Singh Sangher, Amit Singh, Hari Mohan Pandey, and Vinit Kumar. Towards safe cyber practices: Developing a proactive cyber-threat intelligence system for dark web forum content by identifying cybercrimes. *Information*, 14(6), 2023.
- [24] Sayuj Shah et al. MAD-CTI: Cyber threat intelligence analysis of the dark web using a multi-agent framework. *IEEE Access*, 2025.
- [25] Olga E. Sorokoletova et al. Towards a scalable AI-driven framework for data-independent cyber threat intelligence information extraction. In *2024 2nd International Conference on Foundation and Large Language Models (FLLM)*, 2024.
- [26] Harish Thangaraj et al. Cross-lingual transfer of multilingual models on low resource african languages. *arXiv preprint*, 2024.
- [27] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [28] Xuren Wang et al. APTNER: A specific dataset for NER missions in cyber threat intelligence field. In *2022 IEEE 25th International Conference on Computer Supported Cooperative Work in Design (CSCWD)*, 2022.
- [29] Zhen Zeng et al. LICALITY—likelihood and criticality: Vulnerability risk prioritization through logical reasoning and deep learning. *IEEE Transactions on Network and Service Management*, 2022.